



Electrical Resistivity of Anisotropic Sheets



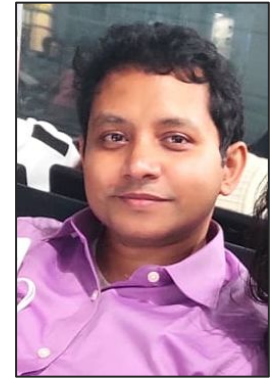
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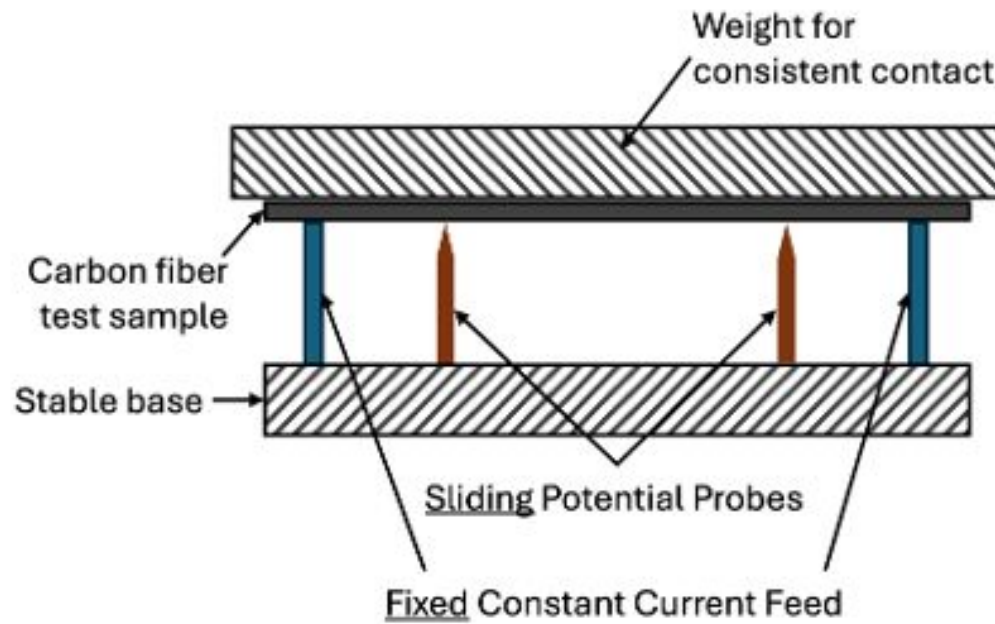
14th October, 2025

DRD8 Collaboration Meeting

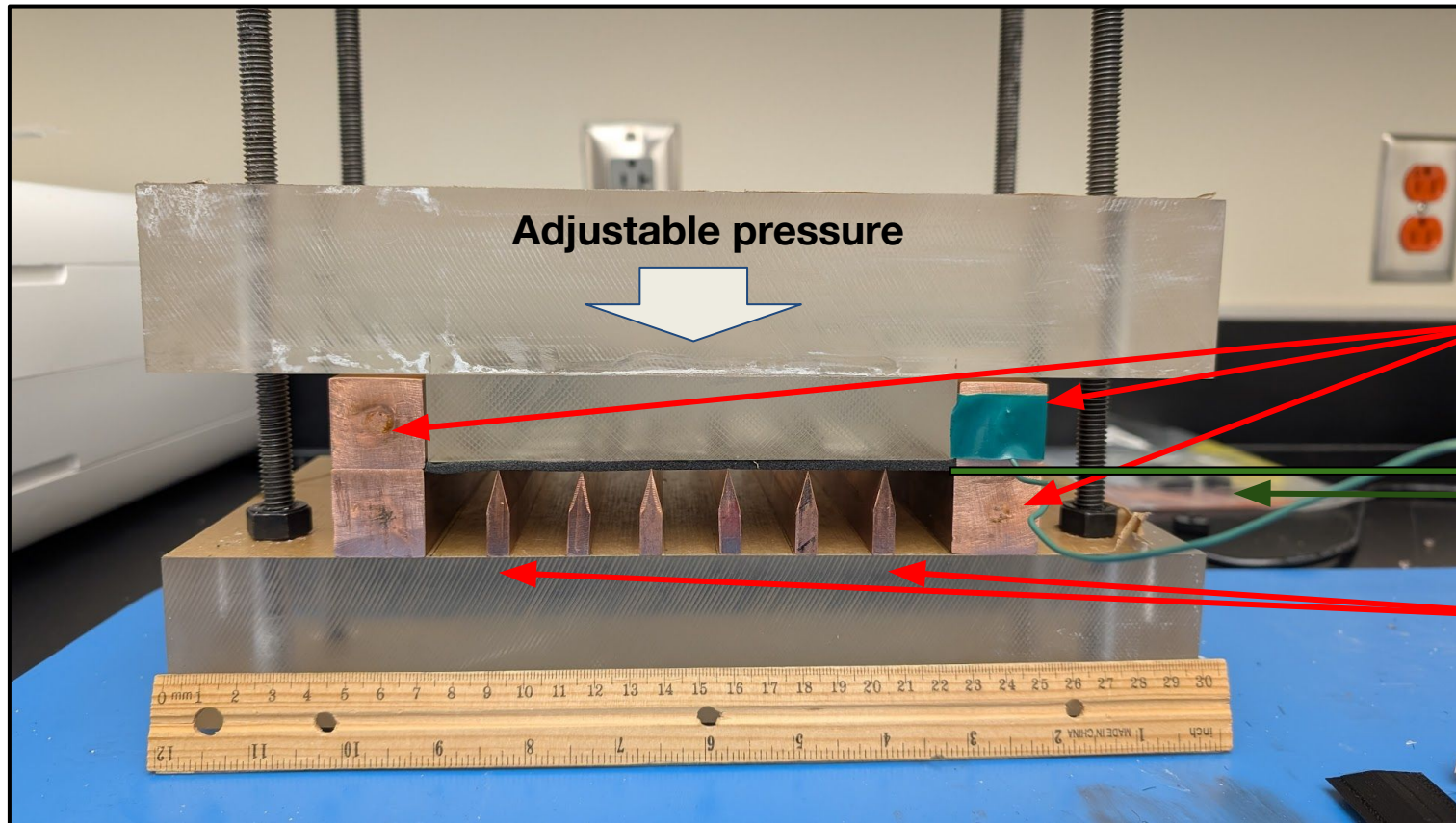
<https://indico.cern.ch/event/1573561/>

- **Electrical Resistivity** measurements of sheet materials
- **Database** for radiation hard materials characterization
 - In collaboration with Nicola Pacifico
 - LLM-based Retrieval Augmented Generation
- **Thermal Conductivity** measurements at steady state

- Electrical resistivity measurements of detector support structures is of paramount importance to understand electrical insulation & grounding
- Materials: carbon fiber (w/ and w/o copper inlays), GEM foils
- Materials can be anisotropic. All components have to be measured.
- Impedance as a function of frequency is relatively unexplored
 - Not sure if we can determine intrinsic properties for impedance, unlike simple resistance ($R = \rho \times \text{length} / \text{area}$)



- Primary proposal was based on **ASTM D4496**
 - But, I think **we can do better** than 2 point probes



Current feeds

**Sample will be
slid in here**

**Knife-edged
voltage probes**

- Four solid copper cuboids serve as current feeds
- 6 sharp copper knife edges serve as potential probes
- Screws used to hold sample in place with adjustable pressure
- Linear fit of potential vs position should eliminate contact resistances
- First calibrations with aluminum sheets will be underway next week